

Aydin Ezzatpour Aghdam

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SUMMARY

Undergraduate Electrical Engineering student at Sharif University of Technology (since 2023; GPA 18.33/20), ranked 119th in Region 1 in the Iranian Nationwide University Entrance Exam. Head/Teaching Assistant experience in engineering mathematics, linear algebra, signals and systems, electronics, and introductory electrical engineering courses. Experienced in recitations, problem-set and quiz design, grading, project/lab support, assignment authoring, and source-backed technical project documentation. Interested in applied mathematics, probability and statistics, linear algebra, machine learning, signal processing, and mathematical modeling.

PROJECTS

Queueing Theory, Markov Chains & Discrete-Event Simulation (Engineering Probability & Statistics)

2024/12

- Analyzed queueing models, Poisson processes, exponential random variables, and Markov chains.
- Implemented reproducible analytical models and continuous-time discrete-event simulations for finite queues, feedback, throughput, and cost optimization.
- Prepared a reviewed English technical report with corrected modeling assumptions and validation against analytical formulas.

License Plate Recognition under Blur (Signals & Systems)

2025/06

- Built a closed-template LPR workflow using grayscale preprocessing, thresholding, contour-based segmentation, and template matching.
- Applied correlation-based recognition, spatial downsampling, blur-direction estimation, and first-order inverse filtering.
- Documented the workflow in a reproducible notebook and LaTeX report with synthetic inputs and saved deblurred outputs.

50 Hz Active Notch-Filter Analysis and Tolerance Study (Circuit Analysis / Electronic Circuit Theory)

2025

- Performed symbolic nodal analysis and transfer-function derivation for a 50 Hz active notch-filter circuit.
- Developed MATLAB scripts for frequency response, Monte Carlo tolerance analysis, component sensitivity, and symbolic expression formatting.
- Prepared an LTspice AC-analysis schematic and documented validation boundaries in an English technical report.

Analog PID Controller Design (Electronics II Laboratory)

2026

- Designed an analog PID controller with proportional, leaky-integral, band-limited derivative, and summing stages.
- Implemented and documented the LTspice schematic using op-amp-based circuit blocks and transient-analysis setup.
- Prepared a technical report and PCB-layout documentation targeting an LM324N quad operational amplifier.

Selective Repeat ARQ and Error-Control Simulation (Data Networks)

2026

- Modeled a bidirectional data-link connection in OMNeT++ using Selective Repeat ARQ.
- Implemented sender/receiver windows, timeouts, acknowledgements, negative acknowledgements, buffering, and retransmission logic.
- Compared parity, Hamming(7,4), and hybrid error-control modes under simulated bit-flip probabilities.

Coding Theory & Decoding (Digital Logic, Dr. Tabandeh)

2025/01

- Studied linear block codes and implemented basic encoder/decoder logic.
- Evaluated error-detection and error-correction trade-offs on synthetic datasets.

HONORS & AWARDS

Iranian Nationwide University Entrance Exam (Konkur)

2023

- Ranked 119th in Region 1 in the highly competitive national university entrance examination.

RESEARCH INTERESTS

Applied Mathematics, Convex Optimization, Machine Learning, Probability and Statistics, Linear Algebra, Signal Processing, Mathematical Modeling

TA EXPERIENCE

Head Teaching Assistant — Engineering Mathematics (Instructor: Dr. Shishegar)

2026/02 — Present

- Served as head TA and coordinated exercise design, quiz design, grading, and recitation-related tasks.
- Designed and authored problem sets and quizzes aligned with the course objectives.
- Conducted problem-solving recitation sessions and supervised grading of assignments and quizzes.

Teaching Assistant — Engineering Mathematics
(Instructor: Dr. Banaei)

2026/02 — Present

- Conducted problem-solving recitation sessions for core topics in engineering mathematics.
- Designed and authored course assignments to strengthen students' conceptual understanding and problem-solving skills.

Teaching Assistant — Linear Algebra
(Instructor: Dr. Molk-Araei)

2026/02 — Present

- Conducted recitation sessions focused on linear algebra concepts and structured problem-solving.
- Graded assignments and exams, providing feedback to support student learning and assessment.

Teaching Assistant — Engineering Mathematics
(Instructor: Dr. Shishegar)

2025/09 — 2026/02

- Designed and authored problem sets covering core topics in engineering mathematics.
- Conducted multiple recitation sessions focused on structured problem-solving techniques.

Teaching Assistant — Signals & Systems (Quizzes & Recitations)
(Instructor: Dr. Behroozi)

2025/09 — 2026/02

- Designed and reviewed course quizzes aligned with learning objectives.
- Held problem-solving sessions to support exam preparation and conceptual understanding.

Project Teaching Assistant — Electronics I
(Project TA Team Member)

2025/09 — 2026/02

- Collaborated with the project TA team on project design and instructional guidelines.
- Assisted students with project implementation, debugging, and technical issues.

Teaching/Project Assistant — Course Projects, Laboratory, and Assignment Design
(Instructor: Dr. Zarghani)

2025/02 — 2025/08

- Designed project-related assignments.
- Designed project course manuals/instructions.
- Assisted in course laboratory sessions and supported student performance evaluation.

Teaching Assistant — Intro to Electrical Engineering
(Instructor: Eng. Jafarpisheh)

2024/09 — 2025/01

- Led problem-solving sessions and clarified fundamental circuit concepts.
- Assisted with quiz preparation and grading.

EDUCATION

B.Sc. in Electrical Engineering
– **Sharif University of Technology**
– GPA: 18.33/20
– Courses of interest: Basic Mathematics, Differential Equations, Engineering Mathematics, Engineering Probability & Statistics, Signals & Systems, Mathematical Methods in Engineering (Linear Algebra), Introduction to Machine Learning, Convex Optimization, Digital Signal Processing, Data Communication Networks

2023 – Present

SKILLS

Programming & Scientific Computing	Python, NumPy, Matplotlib, MATLAB/Simulink, C++
Simulation & Engineering Tools	OMNeT++, PSpice, LTspice, Altium Designer
Document Preparation	L ^A T _E X
Mathematics	Linear Algebra, Differential Equations, Fourier Analysis, Probability and Statistics
Languages	Persian (native), English (good reading/writing)

ACTIVITIES

Scientific Society (Resana)	Active member of the EE student society; contributed to the organization and execution of academic and student events, including logistics and operations.
Academic Workshops	Organized and conducted Engineering Mathematics and Linear Algebra workshops for undergraduate students through Resana.